

A Study on the Generative Art-Based PFP NFT Purchase Journey

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Introduction

Research Objective

- Decision to purchase PFP NFTs based on generative art
- The process by which consumers acquire and utilize relevant information to make decisions and the payment process
- To visualize and understand market entry and repeat consumers



Introduction

Research Necessity

- Growing interest in NFTs among consumers and businesses
- Growth in NFT utilization businesses and channels
- People are increasingly using more and more channels (SNS, media, friends, etc.) exposure to NFTs through
- It is necessary to understand the purchase intention of Profile Picture (PFP) NFTs, which have the highest trading volume among NFTs, from the consumer's perspective.
- Identifying common steps people go through to reach a decision to purchase PFP NFTs
- In the expanding digital environment (WEB 2.0 → WEB 3.0), simplification from the consumer's perspective is needed
- Rethinking the value of NFTs by examining everything from awareness to purchases and PFP NFT use cases.

PFP NFT Definition and Current Status

Non-Fungible Token

- NFTs are a type of cryptocurrency created by smart contracts, tokens linked to assets with different values, characteristics, and different values. (Fairfield, 2021)
- Records of NFT ownership are protected by blockchain, allowing creators to easily prove the existence and ownership of digital assets in the form of videos, images, artworks, event tickets, and more. (Wang et al., 2021)
- Interest and trading volume for NFTs surged significantly in 2020, rising by +21,350% in 2021 to about \$18B. (Nonfungible, 2021)



PFP NFT Definition and Current Status

Profile picture(PFP) NFT

- Among all NFTs, Collectible NFTs ranked first in trading volume (\$8.5B).
- Most of the traded collectible NFTs have been confirmed to be generative art-based NFTs.
- Generative art-based NFTs consist of about 10,000 similar images based on a common template.
- Each image is randomly combined and generated with traits such as race, background color, and various accessories.
- Leading trends by transforming countless celebrities' SNS profile pictures (PFPs) into generative art-based NFTs.
- Celebrities use PFP NFTs on social media as a way to express themselves.



Literature

Consumer Value Theory

Recognized as a key concept for understanding consumer behavior

Consumption Value

- Consumers tend to evaluate the value provided by brands or products and to achieve goals related to their own value by purchasing brands or products that offer special value. (Fandos & Garcia, 2009)
- To explain consumer choice behavior, it integrates research findings from various fields such as economics, sociology, psychology, marketing, and consumer behavior to present the Consumer Value Theory. (Sheth et al, 1991)

Literature

Consumer Value Theory

Recognized as a key concept for understanding consumer behavior

The five types of consumer value

(Sheth et al, 1991)

- Functional value – the physical and practical consumption value related to the price, quality, and functionality of a product
 - Social value – Consumer value related to the social and hierarchical groups that consume products, representing a sense of belonging to the group.
 - Emotional value – the value of consumption associated with negative or positive emotions resulting from product consumption
 - Rare value – Value related to intellectual desire and curiosity, and novelty that drives product consumption
 - Situational value – The value of alternative choices is perceived differently depending on the situation, and is related to product consumption tendencies.
- Consumers are influenced by one or all of the five consumption values during the brand selection process.

Methods

User journey map - Combined analysis of decision models

Research applied to online shopping (Wolny and Charoensuksai, 2014), Research applied to mobile shopping (Grewal et al., 2018) etc, research is increasing mainly

User-Journey Mapping

- A customer-centric methodology that examines the entire process from the customer's awareness stage to purchase and enables integrated analysis (Grewal et al., 2018; Rosenbaum et al., 2017)
- How to graphically describe customer experiences that distinguish interaction touchpoints between products and services of interest – Customer. (Clark, 2013)
- How to Understand Customers in a Multichannel Rapidly Growing Environment (Wolny & Charoensuksai, 2014)
- Helping to Find Efficient Usability Improvement Measures Within the Scope of Usage Scenarios (Jeong Ji-eun & Bang Hwan-hwan, 2014)
- Related research:
 - Types of Mobile Payments (Jung Ji-eun & Bang Hwan-hwan, 2014; Lee Han-jin et al., 2021),
 - Investment Decision Support Design (Jang So-hee & Yoon Jae-young, 2018)

Methods

User journey map - Combined analysis of decision models

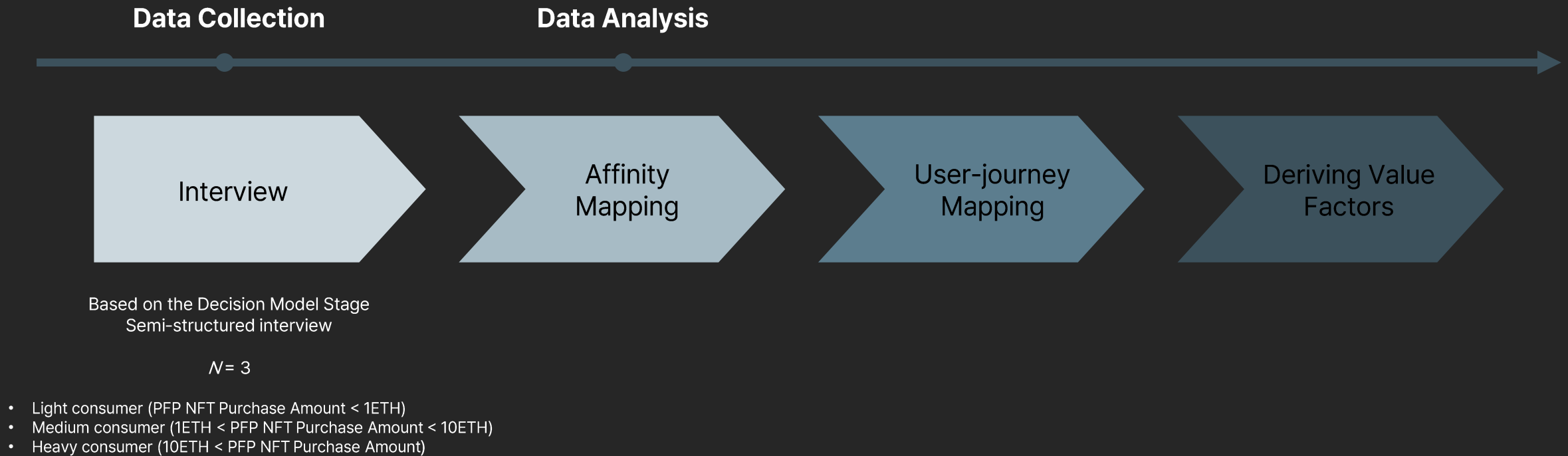
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Decision Model

- Based on the proposition that 'humans are decision-makers who derive the most reasonable and satisfactory results in limited circumstances' (Simon, 1957; Smallman & Moore, 2010)
- Complex compositions based on approaches such as consumer goals, information processing ability, preferences, context, and complexity (Bettman et al., 1998)
- Significantly influenced by various factors such as economic, social, and technological factors (Aldin & de Cesare, 2011)
- Problem Recognition → Exploration, Information Search → Evaluation of Alternatives → Purchase → Post-Purchase (Engel & Roger, 1995; Blackwell, 1982; Lemon & Verhoef, 2016; Puccinelli et al., 2009; Santos & Gonçalves, 2021)

Methods

Research Flow



Result

Derivation and Definition of Consumer Value Factors

Value Category	Experience Factors	Definition
Functional Value	Profitability	Experience to gain profit through the price increase of owned NFTs.
	Online Experiential Elements	Experience to directly experience or utilize within an online environment.
	Offline Experiential Elements	Experience to directly experience or utilize within an offline environment.
	Interoperability	Experience utilizing NFTs in decentralized applications (dApp; DeFi, Metaverse, game, etc.).
	UGC Generation	Motivation to actively create secondary works (UGC) using one's owned NFTs.
Social Value	Conspicuous Consumption	Motivation to show off to others through NFT purchase and utilization.
	Celebrity-Derived Fame	Social effect gained through celebrities who own the same NFT collection.
	Sense of Belonging	Sense of belonging formed through relationship building with other holders of the same NFT collection community.
	Autonomous Participation	Trend of active participation, such as voting or proposing in governance to set problem-solving and direction of the NFT collection.
	Communication	Opportunity for information sharing and interaction through NFT purchase and utilization.
Emotional Value	Positive Emotions	Pleasant emotions gained through NFT ownership and utilization.
	Negative Emotions	Unsatisfied emotions gained through NFT ownership and utilization.
Uniqueness Value	Artistic Taste	Aesthetic satisfaction gained through NFT artwork.
	Differentiation	Trend of wanting to be different from others through purchasing NFTs with high rarity.
	Self-Expression	Self-established individuality through NFT purchase.
	Expectation of New Experiences	Benefits and experiences expected or gained from NFT purchase and utilization.

Result

Key Contents of Customer Journey Map

PFP NFT CUSTOMER JOURNEY MAP					
Phase	Need Recognition Phase	Information Search Phase	Alternative Evaluation Phase	Purchase Area	Post-Purchase Phase
Behavior	Interest in NFT via friend's recommendation	• Search platforms, social media, news • NFT minting sites • NFT communities • Secondary NFT markets	<Between-Project Comparison> • Profitability criteria Check • Taste criteria Check • Popularity criteria Check • Gust criteria Check • Community criteria Check • Expected value-add criteria Check <Within-Project Comparison> • Profitability criteria Check • Differentiation Check • Self-expression possibility Check	• Profitability criteria re-check • Popularity criteria re-check • Expected value-add criteria check	• Community utilisation • Profile picture set • dApps utilisation • UGC utilisation • Governance vote in platform • Info & sharing • Profit generated • Value-add generated
Needs	• Profitability • Principal guarantee • Artwork taste	• Profit guarantee for find information • Risk reduction for find information	<Between-Project Comparison> • Assurance of profitability • Assurance of taste • Assurance of community <Within-Project Comparison> • Assurance of profitability • Desire for conspicuous consumption • Self-expression desire • Assurance of sense of belonging generation	• Re-assurance of profitability • Loyalty to the project of loyalty • Expectation of new experiences	• Profitability on own • UGC utilisation desire • Autonomous participation • Desire for conspicuous consumption • Desire for interaction & sense of belonging • Desire for identity assignment • Desire for value-add • Desire for profitability

Result

Key Contents of Customer Journey Map

Phase	Need Recognition Phase	Information Search Phase	Alternative Evaluation Phase	Purchase Phase	Post-Purchase Phase
Pain Points	- Investment potential - Lack of touchpoints for recognition	- Recognition of complex purchase steps - Fatigue in judging information reliability	- Reliability issue - Monetary/financial issue	Complex purchase steps	- Lack of content and dApps utilizing NFTs - Governance issue - Offline limitations - Added value provision issue - Reliability issue
Experience Factors	Functional Value Profitability	Functional Value Profitability	Functional Value Profitability	Functional Value - Profitability - Artistic Taste	Functional Value - Profitability - Online experiential element - Offline experiential element - Interoperability - UGC generation
	Progressive Value Artistic Taste		Social Value - Conspicuous consumption - Sense of belonging	Social Value Sense of belonging	Social Value - Conspicuous consumption - Sense of belonging - Autonomous participation - Communication
			Emotional Value Positive emotion	Progressive Value Expectation of new experiences	Emotional Value - Positive emotion - Negative emotion
			Progressive Value - Artistic Taste - Differentiation - Self-expression - Expectation of new experiences		Progressive Value - Artistic Taste - Self-expression - Expectation of new experiences

Discussion

Theoretical Significance

- Based on previous research on experiential value, the experiential factors for consumers' PFP NFT purchase decisions were derived.
- Based on the experiences of PFP NFT holders, experiential value factors for each purchase journey were proposed, suggesting that these factors could serve as evidence for future research directions in PFP NFTs.
- Based on this study, the significance of applying consumer value theory to the NFT market, which has high potential for expansion.

Practical significance

- Based on the cognitive motivations, user experiences, and pain points of PFP NFT consumers derived from the combination of decision models (CDM) and consumer journey maps (CDJ), it is suggested that this data can serve as evidence for future PFP NFT issuance and utilization.
- Identify the timeliness of limited PFP NFT consumer research and explore customer behaviors required in the early stages of the NFT market.
- A strategic approach that provides consumer characteristics not only to companies currently issuing and utilizing NFTs but also to those entering the NFT market in the future.

